Nptel Online Certification Course
Indian Institute of Technology Kharagpur
Computer Vision
Assignment - Week 6

Number of questions: 10 **Total marks: 10x2=20**

QUESTION 1: **Type: MCQ**

Given two features each of size 7 as $f(3, 6, 2, 5, 1, 4, 7)$ and $g(3, 4, 3, 1, 2, 3, 9)$. Compute L_1 and L_2 norms between f and g, respectively.

a) 11 and 3.16

b) 11 and 5.20

c) 10 and 5.20

d) 10 and 3.16

Correct Answer: b)
Detailed Solution:
L_1 norm is $\sum_{i=0}^{n} |f - g|$ and L_2 norm is $(\sum_{i=0}^{n} |f - g|^2)^{1/2}$.

QUESTION 2: **Type: MCQ**

Consider the following matrix representing 4 sample data points $(1, 2), (2, 1), (3, 2), (2, 3)$ as $X = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 3 & 2 \\ 2 & 3 \end{bmatrix}$. Compute the covariance matrix using data points from X.

a) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

b) $\begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}$

c) $\begin{bmatrix} 0.25 & 0 \\ 0 & 0.25 \end{bmatrix}$

d) $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$

Correct Answer: b)

Detailed Solution:

Compute the mean vectors as $[\frac{1+2+3+2}{4}, \frac{2+1+2+3}{4}] = [2, 2]$. Now compute

$$\tilde{X} = \begin{bmatrix} 1-2 & 2-2 \\ 2-2 & 1-2 \\ 3-2 & 2-2 \\ 2-2 & 3-2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The covariance matrix is given by $\frac{\tilde{X}^T \tilde{X}}{4}$.

QUESTION 3: **Type: MCQ**

Consider the set of parametric equations $x = 2t^3 - t^2$ and $y = t^3 + 3t$. representing a 2D curve. Compute the tangent vector at $t = 2$.

a) $(4, 15)$

b) $(20, 15)$

c) $(20, 5)$

d) $(15, 5)$

Correct Answer: b)

Detailed Solution:

Perform derivation $x' = 6t^2 - 2t$ and $y' = 3t^2 + 3$. Compute the tangent at $t = 2$.

QUESTION 4: **Type: MSQ**

Which of the following is/are true?

a) RANSAC is best suited for fitting outliers.

b) Hough transform is best suited for line fitting in presence of many lines.

c) Kullback-Leiber Divergence is best suited to find the 'optimal' line parameters when points belonging to a line are given.

d) The average case time complexity of NN Search using K-D tree is $O(log(N) + 2^d)$ where d is the dimension of the feature space.

Correct Answer: b), d)

Detailed Solution:

Kullback-Leibler (KL) Divergence is used to measure the statistical difference between two probability distributions, not geometric distances.

QUESTION 5: **Type: MCQ**

Gradient along x and y directions at a pixel location (x_i, y_i) can be given as $\frac{x_{i+1}-x_{i-1}}{2}$ and $\frac{y_{i+1}-y_{i-1}}{2}$, respectively. The top left corner of the matrix is considered as $(1, 1)$ and indices increment for x and y axes along horizontal right and vertical downward directions, respectively.

Consider a 3×3 matrix $A = \begin{bmatrix} 0.2 & 0.7 & 0.4 \\ 0.4 & 0.2 & 0.9 \\ 0.3 & 0.5 & 0.5 \end{bmatrix}$, compute gradients along x (I_x) and y (I_y) directions, respectively at the center pixel location in A.

a) I_x = -0.05 and I_y = 0.15

b) I_x = 0.05 and I_y = -0.15

c) I_x = 0.25 and I_y = -0.1

d) None of these

Correct Answer: c)

Detailed Solution:

At the center pixel, $I_x = \frac{0.9-0.4}{2}$ and $I_y = \frac{0.5-0.7}{2}$.

QUESTION 6: **Type: MCQ**

If an octave in SIFT operator has 10 images, then the scale factor by which each image in octave differs from the next image is

a) 1.080

b) 1.065

c) 1.071

d) 1.090

Correct Answer: a)

Detailed Solution:

Within an octave, the adjacent scales differ by a constant factor k. If an octave contains $s + 1$ images, then $k = 2^{1/s}$. The scale factor is 2 in an octave. Therefore, $k = 2^{1/9}$.

QUESTIONS 7: **Type: MCQ**

Find the eigen values of the matrix $\begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$.

a) $1, -2$

b) $1 \pm 2i$

c) $-1 \pm 2i$

d) $1, 2$

Correct Answer: b)
Detailed Solution:
Setting up the characteristic determinant:

$$\begin{vmatrix} 1-\lambda & -2 \\ 2 & 1-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)^2 - (-4) = 0$$

$$\lambda^2 - 2\lambda + 1 + 4 = 0$$

$$\lambda^2 - 2\lambda + 5 = 0$$

Using the quadratic formula $\lambda = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$:

$$\lambda = \frac{2 \pm \sqrt{(-2)^2 - 4(1)(5)}}{2} = \frac{2 \pm \sqrt{-16}}{2} = 1 \pm 2i$$

QUESTIONS 8: **Type: MSQ**

Which of the following is/are true for edge and corner detection?

a) A corner will have low maximal and minimal curvatures.

b) A corner will have high maximal and minimal curvatures.

c) An edge will have low maximal curvature, but very high minimal curvature.

d) An edge will have high maximal curvature, but very low minimal curvature.

Correct Answer: b), d)

Detailed Solution:

A corner will have high maximal and minimal curvatures. An edge will have high maximal curvature, but very low minimal curvature.

QUESTION 9: **Type: Numeric**

Consider a 3×3 neighborhood with grey scale pixel values $A = \begin{bmatrix} 80 & 200 & 130 \\ 120 & 192 & 100 \\ 290 & 180 & 132 \end{bmatrix}$, obtain a single Local Binary Pattern (LBP) code for this matrix considering the center pixel as the reference. Weight table to be used is given as $\begin{bmatrix} 2^0 & 2^1 & 2^2 \\ 2^7 & & 2^3 \\ 2^6 & 2^5 & 2^4 \end{bmatrix}$, where top right corner is considered as the first neighbor and this order moves clockwise.

a) 66

b) 64

c) 2

d) 62

Correct Answer: a)

Detailed Solution:

Since the value of center pixel is 192, matrix elements greater than 192 become 1 and 0 otherwise. Hence, after this operation, matrix becomes $\begin{bmatrix} 0 & 1 & 0 \\ 0 & & 0 \\ 1 & 0 & 0 \end{bmatrix}$. Now, multiplying these values with a weight table $\begin{bmatrix} 2^0 & 2^1 & 2^2 \\ 2^7 & & 2^3 \\ 2^6 & 2^5 & 2^4 \end{bmatrix}$ and summing up the corresponding values gives 66.

QUESTION 10: **Type:MSQ**

Which of the following is/are true for SIFT and SURF?

a) Computation of scale space and histograms of gradient direction are approximated in SURF compared to SIFT.

b) Integral of images are used in SURF to speed up computation.

c) Small Gaussians used in SIFT are approximated by box filters in SURF.

d) Haar wavelet responses are used to accumulate the orientation in SIFT and SURF descriptors.

Correct Answer: a), b), c)
Detailed Solution:
